

SUMIT YADAV

AI Researcher | Interpretability & Safety | Applied NLP & Agentic Systems

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SUMMARY

AI researcher and engineer working at the intersection of **mechanistic interpretability, AI alignment/safety, and low-resource NLP**. My research studies how safety-aligned LLMs fail silently — through over-refusals, geometric misrepresentation, and surface-level triggers — and how the internal structure of representations can be understood and steered to make models safer and more reliable; this work has been accepted at **ACL 2026 (CORE A*)**. Alongside research, I design production-grade AI systems: **Zero-Trust** oversight for autonomous agents, **RAG** pipelines over vector databases, self-hosted LLM serving, and security tooling for the **Model Context Protocol**. I am an active Kaggle competitor and open-source contributor (creator of *maiBERT*, the first BERT for Maithili), and I care about extending interpretability and safety to multilingual and underserved communities.

TECHNICAL PROFICIENCY

Interpretability & Alignment: representation & activation steering, mechanistic interpretability, linear probes / CKA, representation geometry, effective dimension, over-refusal mitigation, adversarial probing, insight.

LLMs & NLP: LLMs (Llama 3, Claude, GPT-4o), fine-tuning (PEFT/LoRA), RLHF, RAG pipelines, multi-agent systems, multilingual / low-resource NLP, tokenizer training, embedding alignment.

Computer Vision & Audio: YOLOv8, DINOv2/v3, CLIP, SigLIP, SAM/SAM2, Qwen-VL, Perch / spectrogram models, Mamba hybrids.

AI Security & Agentic Systems: Model Context Protocol (MCP), MITRE ATT&CK mapping, prompt-injection & tool-poisoning defense, Zero-Trust architecture, Guardrails, interactive permission systems.

Architecture: Hexagonal (Ports & Adapters), Domain-Driven Design, microservices, event-driven (SSE), REST/GraphQL APIs.

Vector DBs: Qdrant, ChromaDB, Pinecone, Weaviate (hybrid search, HNSW).

Languages & Tools: Python (expert), C++, C, SQL, Bash, JavaScript; PyTorch, TensorFlow, vLLM, LangChain, LlamaIndex, FastAPI, Docker, Kubernetes, GitHub Actions.

PROFESSIONAL EXPERIENCE

Astha.ai

USA (Remote)

AI Researcher & Architect — Safety & Agentic Systems

May 2025 – Present

- **Zero-Trust agentic oversight:** architected a framework where every agent interaction, tool call, and memory retrieval carries a cryptographic identity and is verified against a strict policy engine, preventing unauthorized actions in multi-agent environments.
- **MCP security proxy:** designed the core MCP-Proxy on a Hexagonal (Ports & Adapters) architecture, decoupling security-policy logic from the SSE transport layer for independent, network-free testing of rules.
- **Policy engine & RBAC:** built a policy engine supporting allow/deny lists (v1) and conditional logic (v2) with Role-Based Access Control (Admin/User/Guest) for granular MCP tool permissions.
- **MCP-Scanner:** led a vulnerability-analysis platform for MCP servers integrating **78+** MITRE ATT&CK-mapped attack techniques, with dual-defense modules for prompt injection (SAFE-T1102) and tool poisoning.

AMNIL Technologies

Lalitpur, Nepal (Hybrid)

AI Engineer — RAG & Infrastructure

Jun 2024 – May 2025

- **Advanced RAG:** built a hybrid-search (sparse + dense) RAG system on **Qdrant** with recursive query decomposition, improving retrieval accuracy by **35%** on multi-hop, domain-specific questions.
- **Self-hosted LLM serving:** deployed open models (Llama 3, Mistral, Qwen) on **vLLM**, tuning PagedAttention / KV-cache to cut latency by **40%** vs. standard Hugging Face pipelines.

- **Guardrails & evaluation:** integrated guardrails to enforce topical boundaries and built an automated LLM-as-a-Judge framework to benchmark response quality.

GradeUp Educations

Kathmandu, Nepal

Data Team Lead

Jan 2022 – Jun 2024

- **Low-resource document AI:** led OCR and document parsing for Nepali and Maithili, overcoming data scarcity via synthetic Devanagari training data; built an automated grade-evaluation system.
- **Semantic search:** fine-tuned embedding models with contrastive learning for educational content, improving relevance for non-English queries; shipped learning agents / chatbots.
- **Leadership:** defined the data roadmap and led a small team across the ML lifecycle from ideation to deployment.

DeepLearning.AI

Global (Remote)

GAN Specialization Mentor

Aug 2021 – Present

- Technical mentor for the Generative Adversarial Networks Specialization — loss functions (Minimax, Wasserstein), training stability, and architectures (DCGAN, CycleGAN) — supporting students worldwide.

RESEARCH & PUBLICATIONS

Peer-reviewed and preprint work at the intersection of AI safety, interpretability, and low-resource NLP. Full list on Google Scholar.

SafeConstellations: Mitigating Over-Refusals in LLMs Through Task-Aware Representation Steering | *AI Safety, Representation Steering*

- LLMs trace consistent, task-specific “constellation” trajectories in embedding space that shift predictably between refusal and non-refusal states.
- Proposed an inference-time, task-aware trajectory-shifting method that reduces over-refusals by up to **73%** with minimal utility loss and no retraining — accepted at **ACL 2026 Main Conference**. (with U. Maskey, M. Dras, U. Naseem.) [\[ACL Anthology\]](#) [\[arXiv\]](#)

Geometric Phases of Mechanism Formation in Neural Networks (sole-authored) | *Interpretability, Training Dynamics*

- Using linear probes and CKA across dense training checkpoints, shows classification mechanisms form **output-layer-first** within the first ~5% of training (Cohen’s $d = 3.68$); the deep-first pattern holds in from-scratch LLM pretraining (GPT-2 Small, SmolLM2-135M) and reproduces on public Pythia and OLMo-2 checkpoints.

On the Relationship Between Representation Geometry and Generalization (sole-authored) | *Representation Learning, Information Geometry*

Pre-Print 2026

- Establishes that a label-free geometric metric — **effective dimension** — strongly predicts generalization across vision and language (partial $r = 0.75$ over 52 ImageNet models, 13 architectures; carries to NLP encoders and decoder-only LLMs).
- Shows the link is causal: degrading the geometry hurts accuracy while PCA cleanup preserves it — a diagnostic relevant to distribution-shift detection and robustness. [\[arXiv\]](#)

MaiBERT: A Pre-training Corpus and Language Model for Low-Resourced Maithili | *NLP, Low-Resource Languages*

LoResLM @ ACL 2026

- First monolingual BERT for Maithili (~50M speakers), pre-trained on a custom-curated corpus with a Masked Language Modeling objective.
- Achieves **87.02%** news-classification accuracy, outperforming multilingual baselines (MuRIL, NepBERTa) — addressing fairness and access for underrepresented communities. [\[ACL Anthology\]](#) [\[Hugging Face\]](#)

Steering Hindi Competence into Unsupported Devanagari Languages | *Multilingual Steering*

Working Paper 2026

- With **no fine-tuning**, steers a model’s Hindi representations to elicit related Devanagari languages it was never trained on: on Llama-3.1-8B and Aya-23-8B a single steering direction produces **genuine Nepali**, with Hindi outperforming English as the source because it sits closer in representation space.

Revolutionizing Currency Security with YOLOv8 | *Computer Vision, Object Detection* J. Bus. Econ. Stud. 2024

- Applied YOLOv8 to counterfeit Nepali banknote detection on a custom dataset, achieving **0.986** true-positive recall on the reverse face — enabling a practical mobile verification solution. [\[DOI\]](#)

Machine Learning Analysis of Tirhuta Lipi | *Script Recognition, Cultural Heritage* 2023

- Character-level analysis and recognition for the historic Tirhuta (Mithilakshar) script, building the data and modeling groundwork for digitizing an endangered writing system.

Support Vectors for Imbalanced Text Classification | *Machine Learning Theory* 2023

- Showed that for highly imbalanced datasets with 100+ classes, optimized SVMs with TF-IDF often outperform deep models on training efficiency and F1-score stability.

COMPETITIVE ML & OPEN SOURCE

Kaggle & Data-Science Competitions (2025–2026) | *20+ models across 5 challenges* Global

- **BirdCLEF+ 2026** — acoustic species identification from Pantanal wildlife audio. Trained DINOv2-based spectrogram classifiers (224px), convolutional baselines, and Google Perch 2.0 transfer learning; earned a **Bronze medal** with a kNN-probe on Perch embeddings.
- **CSIRO Image2Biomass** — pasture above-ground biomass estimation from images + metadata. Built hybrid vision models: DINOv3 + Mamba (3-head, improved, SigLIP-Mamba variants), DINOv3 + SAM, and DINOv2 baselines.
- **Vesuvius Challenge (Ink Detection)** — 2.5D/3D ink segmentation on carbonized Herculaneum scrolls. Explored DINOv2 2.5D, 3D UNet variants with topological features (multi-GPU), and TransUNet + DINOv2 ensembles.
- **Scientific Image Forgery Detection (Recod.ai/LUC)** — copy-move detection & segmentation in biomedical images. Broad architecture sweep: CNN–DINOv2 hybrids, ConvNeXt + progressive decoders, EfficientNet-B0, YOLO + SAHI, fine-tuned SAM2 / Qwen3-VL, and steered vision models across multiple ensembles.
- **Community:** founder of **NPL Coders**, organizing national-level data-science hackathons to grow Nepal's AI ecosystem.

Open Source | *GitHub: rockerritesh* Open Source

- **maiBERT-TF** — first open BERT pre-trained for Maithili, used by the linguistic-research community. **Maithili speech** — open Whisper-tiny ASR and MMS-TTS TTS with live HF Spaces demos.
- **SAFE-MCP** — maintainer of the community security framework for the Model Context Protocol; authored ATT&CK-style detection rules and PoCs (server enumeration, tool-chaining pivots, multimodal prompt injection).
- **Telebot-Claude-Bridge** — permission-gated, human-in-the-loop remote control of the Claude Code CLI via Telegram.

KEY ENGINEERING PROJECTS

Telebot-Claude-Bridge | *Agentic tooling, Python + Telegram API* 2025–2026

- Production bridge for remote control of the **Claude Code CLI** entirely from Telegram: bot ↔ HTTP webhook ↔ tmux-hosted session with hooks relaying tool calls, stops, and permission prompts in real time.
- Interactive Allow / Deny / Allow-All permission buttons per tool call (zero-trust), multi-session switching with auto-recovery, and code-block-aware message splitting. [\[GitHub\]](#)

Nepali Chat-with-Doc & Localization Suite | *RAG, Unicode, Font Conversion* Production

- Full-stack RAG chatbot optimized for Nepali, with a real-time Preeti↔Unicode conversion layer that lets the LLM process legacy government documents; reflexive agents self-correct on poor retrieval.

- Agents.ai & Semantic Router | Vector DB, Multi-Agent Systems2024
 - Intent-aware router that selects the best expert agent (research / coding / data) by semantically matching queries against capability profiles stored as embeddings in ChromaDB.
- IRB Robotic Arm | IoT, Signal Processing, C++2020
 - Image-Recognition-Based robotic arm for touchless medical assistance (UN SDG3): TensorFlow gesture recognition with Arduino actuator control.

EDUCATION

- Pulchowk Engineering College, IOE — Tribhuvan UniversityKathmandu, Nepal
 - Bachelor of Computer Engineering2019 – 2024
 - B.E. thesis: “Evaluating Auto-Encoding Transformer Language Models for Maithili Text Classification” — my first Maithili masked language model and the foundation for maiBERT.
 - Key subjects: Artificial Intelligence, Data Structures & Algorithms, Distributed Systems, Big Data Technologies, Network Security, Compiler Design.

SERVICE, MENTORSHIP & COMMUNITY

- Reviewer:** Mechanistic Interpretability Workshop, ICML 2026 (recognized as an **Outstanding Reviewer**).
- Mentorship:** technical mentor, DeepLearning.AI GANs Specialization (2021–present), supporting students worldwide.
- Community building:** founder of **NPL Coders**, organizing national data-science hackathons to grow Nepal’s AI ecosystem; maintainer and contributor across open-source NLP and AI-security projects.
- Open science:** public releases of maiBERT, Maithili ASR/TTS, and interpretability experiments with reproducible code and live demos.

HONORS & AWARDS

- Outstanding Reviewer**, Mechanistic Interpretability Workshop (ICML 2026) · **Winner**, GritFeat AI Hackathon (2023) · **1st Runner-Up**, Locus Dataverse (2023) · **1st Runner-Up**, Docsumo DataRush (2022)
- Best AI Project**, DELTA 3.0 (2023) · **Winner**, IT-Meet Image Challenge (2022) · **Winner**, LogPoint Capture The Flag (2022)

SELECTED CERTIFICATIONS

- Deep Learning Specialization — DeepLearning.AI (Coursera).
- Generative Adversarial Networks (GANs) Specialization — DeepLearning.AI (Coursera).

LANGUAGES

- Maithili (native) · Nepali (fluent) · Hindi (fluent) · English (professional working proficiency).